

Concomitant Treatment of Malignant Brain Tumours With CBD - A Case Series and Review of the Literature.

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Grade IV glioblastoma multiforme is a deadly disease, with a median survival of around 14 to 16 months. Maximal resection followed by adjuvant radiochemotherapy has been the mainstay of treatment since many years, although survival is only extended by a few months. In recent years, an increasing number of data from in vitro and in vivo research with cannabinoids, particularly with the non-intoxicating cannabidiol (CBD), point to their potential role as tumour-inhibiting agents. Herein, a total of nine consecutive patients with brain tumours are described as case series; all patients received CBD in a daily dose of 400 mg concomitantly to the standard therapeutic procedure of maximal resection followed by radiochemotherapy. By the time of the submission of this article, all but one patient are still alive with a mean survival time of 22.3 months (range=7-47 months). This is longer than what would have been expected.

Revolutionizing Glioblastoma Treatment: A Comprehensive Overview of Modern Therapeutic Approaches.

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Glioblastoma is the most common malignant primary brain tumor in the adult population, with an average survival of 12.1 to 14.6 months. The standard treatment, combining surgery, radiotherapy, and chemotherapy, is not as efficient as we would like. However, the current possibilities are no longer limited to the standard therapies due to rapid advancements in biotechnology. New methods enable a more precise approach by targeting individual cells and antigens to overcome cancer. For the treatment of glioblastoma, these are gamma knife therapy, proton beam therapy, tumor-treating fields, EGFR and VEGF inhibitors, multiple RTKs inhibitors, and PI3K pathway inhibitors. In addition, the increasing understanding of the role of the immune system in tumorigenesis and the ability to identify tumor-specific antigens helped to develop immunotherapies targeting GBM and immune cells, including CAR-T, CAR-NK cells, dendritic cells, and immune checkpoint inhibitors. Each of the described methods has its advantages and disadvantages and faces problems, such as the inefficient crossing of the blood-brain barrier, various neurological and systemic side effects, and the escape mechanism of the tumor. This work aims to present the current modern treatments of glioblastoma.

Unusual presentation of glioblastoma in the brainstem: a case report of a diffuse pontine glioblastoma multiforme and surgical management.

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Diffuse pontine glioblastoma multiforme is a rare subtype of glioblastoma associated with a poor prognosis. In this case report, we present a unique case of diffuse primary pontine glioblastoma multiforme in a patient without any supratentorial lesions. We review the symptoms, treatment options, and case management of patients with infratentorial glioblastoma multiforme and compare these with our patient. Our patient presented with symptoms including progressive diplopia, gait disturbance, and lower extremity weakness. Magnetic resonance imaging revealed a diffuse lesion involving the pons and

biopsy revealed only mildly-atypical glial infiltrates. Consequentially, diagnosis was driven by genetic analysis. Due to the location of the tumor, surgery was not considered a viable option. Instead, the patient received radiation therapy along with concomitant and adjuvant temozolomide chemotherapy which has resulted in improvement of symptoms. This case highlights the challenges of managing diffuse primary pontine glioblastoma multiforme and the need for more effective treatment options for this rare subtype of glioblastoma. Despite aggressive treatment, the prognosis for patients with infratentorial glioblastoma multiforme remains poor, with a median survival time of less than a year. Further research is needed to improve our understanding of the biology and optimal management of this disease.

Characterizing Cell Stress and GRP78 in Glioma to Enhance Tumor Treatment.

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Glioblastoma (GBM) is the most common primary brain tumor, carrying a very poor prognosis, with median overall survival at about 12 to 15 months despite surgical resection, chemotherapy with temozolomide (TMZ), and radiation therapy. GBM recurs in the vast majority of patients, with recurrent tumors commonly displaying increase in resistance to standard of care chemotherapy, TMZ, as well as radiotherapy. One of the most commonly cited mechanisms of chemotherapeutic and radio-resistance occurs via the glucose-regulated protein 78 (GRP78), a well-studied mediator of the unfolded protein response (UPR), that has also demonstrated potential as a biomarker in GBM. Overexpression of GRP78 has been directly correlated with malignant tumor characteristics, including higher tumor grade, cellular proliferation, migration, invasion, poorer responses to TMZ and radiation therapy, and poorer patient outcomes. GRP78 expression is also higher in GBM tumor cells upon recurrence. Meanwhile, knockdown or suppression of GRP78 has been shown to sensitize cells to TMZ and radiation therapy. In light of these findings, various novel developing therapies are targeting GRP78 as monotherapies, combination therapies that enhance the effects of TMZ and radiation therapy, and as treatment delivery modalities. In this review, we delineate the mechanisms by which GRP78 has been noted to specifically modulate glioblastoma behavior and discuss current developing therapies involving GRP78 in GBM. While further research is necessary to translate these developing therapies into clinical settings, GRP78-based therapies hold promise in improving current standard-of-care GBM therapy and may ultimately lead to improved patient outcomes.

Serum lactate dehydrogenase as a prognostic marker for treatment response in IDH wild-type glioblastoma patients undergoing stupp protocol.

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Elevated lactate dehydrogenase (LDH), a marker of tumor aggressiveness and metabolic alterations, may predict treatment response and overall survival across various tumors. This study investigates the correlation between serum LDH levels and clinical outcomes in glioblastoma patients treated with radiotherapy (RT) and temozolomide (TMZ). This retrospective study analysed patients with IDH wild-type glioblastoma (IDH-wt GB) treated at the Radiotherapy Department of Azienda Ospedaliero-Universitaria Senese from 2018 to 2023. Clinical data, including hematologic parameters (e.g., LDH), imaging (MRI), and MGMT promoter methylation status, were collected. All patients received RT and TMZ following the Stupp protocol. Serum LDH levels were measured one week before RT, and Radiological Response (RR)

was assessed using RANO criteria. Overall survival (OS), progression-free survival (PFS), and RR were primary endpoints. Statistical analyses included Kaplan-Meier, Cox regression, and decision tree analysis for LDH cut-off determination. In a cohort of 147 IDH wild-type glioblastoma patients treated with the Stupp protocol, the median OS was 14 months and median PFS was 8 months. Elevated baseline LDH levels were associated with significantly poorer outcomes, showing a median OS of 9 months versus 20 months and a median PFS of 6 months versus 13 months for lower LDH levels ($p < 0.001$ and $p = 0.0001$, respectively). LDH levels also correlated with RR ($p = 0.001$). Multivariate analysis confirmed high LDH as an independent predictor of worse OS ($HR = 2.31$) and PFS ($HR = 2.60$), suggesting its utility as a prognostic biomarker. Elevated LDH levels before starting the Stupp protocol are clinically significant as they predict poorer overall survival and progression-free survival in glioblastoma patients and worse RR. Incorporating LDH measurements into treatment planning can help identify patients at higher risk of poor outcomes, allowing for more tailored and potentially aggressive treatment strategies to improve management and therapeutic responses in glioblastoma.

Enhancing Glioblastoma Resection with NIR Fluorescence Imaging: A Systematic Review.

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Glioblastoma (GB) is among the most aggressive and difficult-to-treat brain tumors, with a median survival of only 12-15 months despite maximal treatments, including surgery, radiotherapy, and chemotherapy. Extensive surgical resection improves survival in glioblastoma patients; however, achieving complete resection is often hindered by limitations in neurosurgical guidance technologies for accurate tumor margin detection. Recent advancements in fluorescence-guided surgery (FGS) and imaging techniques have significantly enhanced the precision and extent of glioblastoma resections. This study evaluates the impact of NIR fluorescence imaging on tumor visualization, surgical precision, cost-effectiveness, and patient survival. A systematic review of PubMed, Scopus, Google Scholar, and Embase was conducted to identify studies on the role of NIR fluorescence in glioblastoma surgery. A total of 135 studies were included, comprising 10 reviews, three clinical studies, 10 randomized controlled trials (RCTs), 10 preclinical studies, and four case reports, all focused on NIR fluorescence imaging in glioblastoma surgery. The findings indicate that NIR fluorescence imaging significantly improves tumor visualization, resulting in an 18-22% increase in gross total resection (GTR) rates in clinical studies. NIR fluorescence provides continuous real-time feedback, minimizing repeat imaging, reducing operational costs, and increasing GTR. These improvements contribute to better patient outcomes, including extended progression-free survival, improved overall survival, and reduced postoperative neurological deficits. This review underscores the potential of NIR imaging to establish a new standard for intraoperative glioblastoma management.

Pharmacological Blockade of Group II Metabotropic Glutamate Receptors Reduces the Incidence of Brain Tumors Induced by Prenatal Exposure to N-ethyl-N-nitrosourea in Rats.

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The study demonstrates that pharmacological blockade of type 3 metabotropic glutamate (mGlu3)

receptors at the time of tumor induction significantly reduces the incidence of brain gliomas in rats. The overall survival of patients with high-grade brain gliomas is 14-20 months after current multimodal therapy, including surgery, radiotherapy, and adjuvant chemotherapy. To demonstrate in this experimental model that pharmacological blockade of group II metabotropic glutamate receptors reduces the incidence of brain tumors induced by prenatal exposure to N-ethyl-N-nitrosourea (ENU) in rats. Dams received a single injection of ENU (40 mg/kg, e.v.) at day 20 of pregnancy, combined with 5 daily injections of either saline or the mGlu2/3 receptor antagonist, LY341495 (10 mg/kg) (from day 15 to day 21 of pregnancy). Assessment of brain tumors in the offspring at 5 months of age showed the presence of mixed gliomas (astrocytomas/oligodendrogliomas) in 70% of the ENU + saline group of rats and only in 30% of the ENU + LY341495 group. Tumors in both groups of rats showed a moderate/high expression of the astrocyte marker, GFAP, and the oligodendrocyte marker, OLIG-2, and a low expression of the proliferation marker, Ki-67. However, tumors of the ENU + LY341495 group showed a reduced density of Iba-1+ cells, suggesting a lower extent of neuroinflammation in the tumor microenvironment. These findings strengthen the hypothesis that mGlu3 receptors are candidate drug targets for the treatment of malignant gliomas.

Radiosensitizing capacity of fenofibrate in glioblastoma cells depends on lipid metabolism.

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Despite advances in multimodal therapy approaches such as resection, chemotherapy and radiotherapy, the overall survival of patients with grade 4 glioblastoma (GBM) remains extremely poor (average survival time <2 years). Altered lipid metabolism, which increases fatty acid synthesis and thereby contributes to radioresistance in GBM, is a hallmark of cancer. Therefore, we explored the radiosensitizing effect of the clinically approved, lipid-lowering drug fenofibrate (FF) in different GBM cell lines (U87, LN18). Interestingly, FF (50 μ M) significantly radiosensitizes U87 cells by inducing DNA double-strand breaks through oxidative stress and impairing mitochondrial membrane integrity, but radioprotects LN18 cells by reducing the production of reactive oxygen species (ROS) and stabilizing the mitochondrial membrane potential. A comparative protein and lipid analysis revealed striking differences in the two GBM cell lines: LN18 cells exhibited a significantly higher membrane expression density of the fatty acid (FA) cluster protein transporter CD36 than U87 cells, a higher expression of glycerol-3-phosphate acyltransferase 4 (GPAT4) which supports the production of large lipid droplets (LDs), and a lower expression of diacylglycerol O-acyltransferase 1 (DGAT1) which regulates the formation of small LDs. Consequently, large LDs are predominantly found in LN18 cells, whereas small LDs are found in U87 cells. After a combined treatment of FF and irradiation, the number of large LDs significantly increased in radioresistant LN18 cells, whereas the number of small LDs decreased in radiosensitive U87 cells. The radioprotective effect of FF in LN18 cells could be associated with the presence of large LDs, which act as a sink for the lipophilic drug FF. To prevent uptake of FF by large LDs and to ameliorate its function as a radiosensitizer, FF was encapsulated in biomimetic cell membrane extracellular lipid vesicles (CmEVs) which alter the intracellular trafficking of the drug. In contrast to the free drug, CmEV-encapsulated FF was predominantly enriched in the lysosomal compartment, causing necrosis by impairing lysosomal membrane integrity. Since the stability of plasma and lysosomal membranes is maintained by the presence of the stress-inducible heat shock protein 70 (Hsp70) which has a strong affinity to tumor-specific glycosphingolipids, necrosis occurs predominantly in LN18 cells having a lower membrane Hsp70 expression density than U87 cells. In summary, our findings indicate that the lipid metabolism of tumor cells can affect the radiosensitizing capacity of FF when encountered

either as a free drug or as a drug loaded in biomimetic lipid vesicles.

CXCL12 impact on glioblastoma cells behaviors under dynamic culture conditions: Insights for developing new therapeutic approaches.

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Glioblastoma multiforme (GBM) is the most prevalent malignant brain tumor, with an average survival time of 14 to 20 months. Its capacity to invade brain parenchyma leads to the failure of conventional treatments and subsequent tumor recurrence. Recent studies have explored new therapeutic strategies using a chemoattracting gradient to attract GBM cells into a soft hydrogel trap where they can be exposed to higher doses of radiation or chemotherapy. It has been demonstrated in vitro under static conditions, that nanoparticles (NPs) encapsulating the chemoattractant CXCL12 can create a gradient to attract GBM cell. However, GBM cell invasion is also largely dependent on interstitial fluid flow (IFF). In the present study, a custom-made in vitro 3D model with indirect perfusion to mimic IFF at flow rates of 0.5 $\mu\text{L}/\text{min}$ and 3 $\mu\text{L}/\text{min}$ was used to examine the invasive behavior of F98-rodent-derived and U87-human-derived GBM cells. This model simulated IFF and CXCL12 gradient within an alginate:matrigel-based hydrogel mimicking brain parenchyma. Findings revealed that CXCL12 (1600 ng/mL) released from NPs significantly increased the migration of F98 GBM cells after 72 hours under IFF conditions at both 0.5 and 3 $\mu\text{L}/\text{min}$. In contrast, U87 GBM cells required a higher CXCL12 concentration (2400 ng/mL) and longer incubation time for migration (120 hours). Unlike the F98 cells, U87 GBM cells showed a CXCL12 dose-dependent proliferation. Semi-quantitative qPCR showed higher CXCR4 mRNA levels in F98 cells than in U87 cells. CXCL12 significantly increased intracellular calcium levels via CXCR4 activation, with a 2.3-fold rise in F98 cells compared to U87, consistent with observed cell behavior during perfusion. This highlights the combined influence of IFF and CXCL12 on cell migration, dependent on cell line. This 3D dynamic model is a valuable tool to analyze parameters like interstitial fluid flow (IFF) and chemokine gradients, influenced by GBM tumor diversity.

Supra-complete surgery via dual intraoperative visualization approach (DiVA) prolongs patient survival in glioblastoma.

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Safe and complete resection represents the first step in the treatment of glioblastomas and is mandatory in increasing the effectiveness of adjuvant therapy to prolong overall survival. With gross total resection currently limited in extent to MRI contrast enhancing areas, the extent to which supra-complete resection beyond obvious contrast enhancement could have impact on overall survival remains unclear. DiVA (dual intraoperative visualization approach) redefines gross total resection as currently accepted by enabling for the first time supra-complete surgery without compromising patient safety. This approach exploits the advantages of two already accepted surgical techniques combining intraoperative MRI with integrated functional neuronavigation and 5-ALA by integrating them into a single surgical approach. We investigated whether this technique has impact on overall outcome in GBM patients. 105 patients with GBM were included. We achieved complete resection with intraoperative MRI alone according to current best-practice in glioma surgery in 75 patients. 30 patients received surgery with supra-complete resection. The control arm showed a median life expectancy of 14 months, reflecting current

standards-of-care and outcome. In contrast, patients receiving supra-complete surgery displayed significant increase in median survival time to 18.5 months with overall survival time correlating directly with extent of supra-complete resection. This extension of overall survival did not come at the cost of neurological deterioration. We show for the first time that supra-complete glioma surgery leads to significant prolongation of overall survival time in GBM patients.
